Three years outdoor exposure of low carbon steel in Mauritius

Baboo Y.R. Surnam
Department of Mechanical and Production Engineering, University of Mauritius, Reduit, Mauritius

Abstract

Purpose – This paper aims to investigate the corrosion behaviour of carbon steel in the Mauritian atmosphere over a three-year period. Atmospheric corrosion is a serious problem in Mauritius.

Design/methodology/approach – Carbon steel samples were exposed outdoors at various sites. Mass loss analysis was performed to determine the corrosion behaviour of the metal over the exposure period. Scanning electron microscopy and Raman tests were performed to investigate the formation of the corrosion products on the carbon steel surface.

Findings – It was found that the corrosion loss at two of the sites considered did not vary clearly according to the bilogarithmic law. Time of wetness was found to be a main factor affecting atmospheric corrosion in Mauritius. The corrosivity of the atmosphere was found to lie between categories C3 and C4, according to ISO 9223.

Originality/value – The results can be of essential help to the construction industry, especially as steel buildings are becoming very common in Mauritius. Moreover, as Mauritius is a tropical island, the results obtained can be useful in other tropical islands.

Keywords Modelling and prediction, Atmospheric

Paper type Research paper

Introduction

In Mauritius, imports of iron and steel increased from 46,000 tonnes in 2000 to 68,200 tonnes in 2007 (Central Statistical Office, 2011). Imports have now increased to over 110,697 tonnes per year. These metals are prone to atmospheric corrosion attack and their use, especially in the construction of buildings and other structures, can incur substantial corrosion costs. Moreover, corroded steel structures eventually end up as waste. If they are not recycled, they can create environmental and social problems. World-wide, studies have shown that the overall cost of corrosion amounts to at least 4-5 per cent of the gross national product, and 20-25 per cent of this cost could be avoided by using appropriate corrosion control technology. Atmospheric corrosion is a major contributor to this cost (Castaño *et al.*, 2010; Corvo *et al.*, 2008).

Several studies have been undertaken in this field in many countries worldwide. It has generally been observed that rural and urban regions have lower corrosion rates than marine and industrial sites (de la Fuente *et al.*, 2011). This is due to the fact that airborne salinity (Cole *et al.*, 2003; Santana *et al.*, 2001; Singh *et al.*, 2008) in marine sites and sulphur dioxide level (Bhaskar *et al.*, 2004; Lan *et al.*, 2006) in industrial sites have been found to influence significantly the rate of atmospheric corrosion. Time of wetness (TOW), which depends on the relative humidity, is considered as another factor affecting atmospheric corrosion, especially in tropical humid climates (Veleva and Maldonado, 1998; Cai and Lyon, 2005). In other studies, it was observed that sulphur dioxide, airborne salinity and relative humidity are the major factors affecting atmospheric corrosion (Roberge *et al.*, 2002; Natesan *et al.*, 2006). Furthermore, the International Organisation for Standardisation (ISO) classification system for the corrosivity of atmospheres takes into consideration TOW, sulphur dioxide content and airborne salinity (ISO 9223, 1992) as the main factors used in the categorisation of atmospheric corrosivity.

In tropical countries, it has been often observed that in rural regions, the corrosivity category is in the range C2 to C3, according to ISO 9223, while in marine and industrial atmospheres, the corrosivity generally falls in category C4 to C5 (Castaño *et al.*, 2010; Corvo *et al.*, 2008; Lan *et al.*, 2006; Veleva and Maldonado, 1998). Mauritius, being a tropical island, is expected to have similar corrosion behaviour.

Mauritius is a small island of area approximately 1,865 km^2. It lies about 800 km east of Madagascar, as shown in Figure 1. It has abundant rainfall, ranging from 900 mm per year to 5,000 mm per year, with an average of 2,000 mm per year overall, and relative humidity is frequently above 80 per cent. Being surrounded by sea and having substantial rainfall and high relative humidity, the level of corrosivity of the Mauritian atmosphere is expected to be high.

Hence, the aim of this study was to get a better understanding of the atmospheric corrosion behaviour of carbon steel in the Mauritian atmosphere through outdoor exposure of the metal over a three-year period. This was

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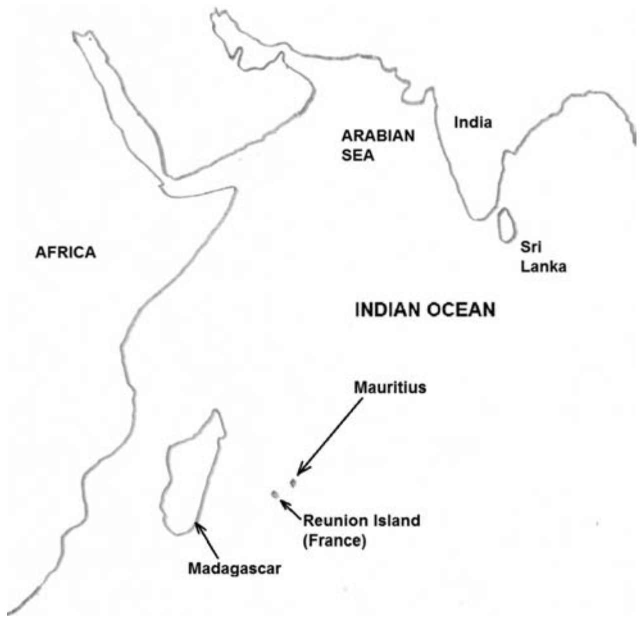
Anti-Corrosion Methods and Materials
62/4 (2015) 246–252

[DOI 10.1108/ACMM-12-2013-1328]

Received 3 December 2013
Revised 24 May 2014
Accepted 7 October 2014

Figure 1 Location of Mauritius in the Indian Ocean



expected to enable engineers to estimate corrosion allowances for steel structures with greater precision and also to allow corrosion management strategies to be improved in the industry.

Methodology

Carbon steel, of composition as shown in Table I, was exposed outdoors for three years according to BS EN ISO 8565 (1995), as shown in Figure 2. The test sites chosen were Reduit, Port Louis, St Julien d'Hotman and Belle Mare, as shown in Figure 3 and as described below:

- *Reduit*: A rural region, 9 km from the nearest coast and very near to industrial zones and the other towns.
- *Belle Mare*: A site situated on the eastern coast of Mauritius, 50 m from the shoreline. It is a purely marine site, far away from industrial and urban regions.
- *Port Louis*: The capital of Mauritius and the site is found at the shoreline and in an industrial zone. Therefore, it can be considered as a marine and industrial region.
- *St Julien d'Hotman*: This site is found far away from the industrial and urban regions and can be regarded as a rural site 16 km inland.

Table I Composition of the main alloying elements

Alloying elements	Weight (%)
Carbon	0.246
Sulphur	0.0269
Manganese	0.238
Phosphorus	0.000
Silicon	0.004
Nickel	0.024
Chromium	0.026
Copper	0.033

Figure 2 Exposure of low carbon steel samples according to BS EN ISO 8565

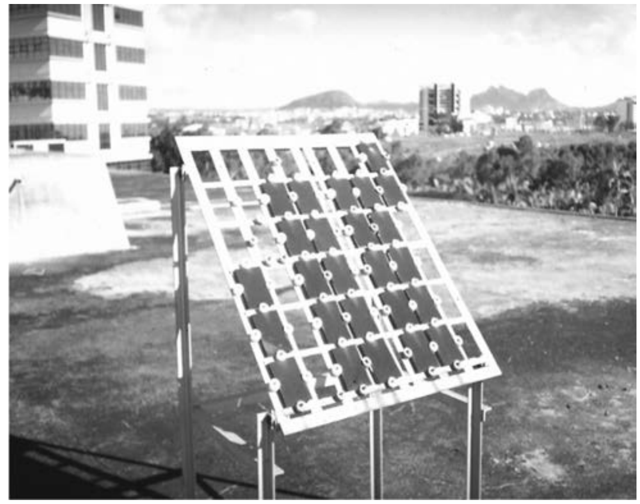
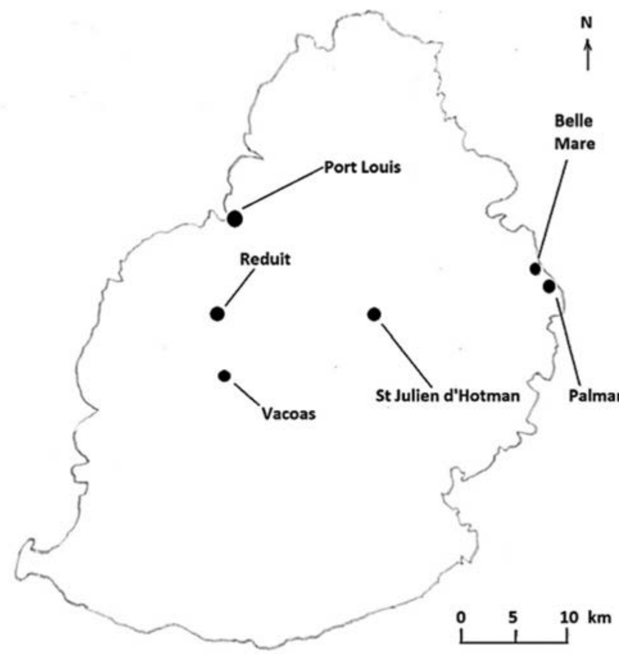


Figure 3 Location of the sites chosen



The samples were removed from the exposure racks, for mass loss analysis, after approximate time intervals of 3, 6, 12, 18 and 38 months in sets of four. Prior to the weight loss determinations, scanning electron microscopy (SEM) and Raman tests were performed on the corroded sample surfaces. The mass loss analysis consisted of cleaning the metal samples, removing the corrosion products, noting the mass loss and determining the corrosion loss of the samples. The corrosion products on the metal surfaces were removed by making use of chemicals according to BS 7545 (1991). Prior to this, a light mechanical cleaning treatment by brushing with a soft bristle brush under running water was first applied to remove lightly adherent corrosion products. The mass loss for each sample was determined for all the four samples removed.

Table II Atmospheric parameters considered at the sites

Site	Average chloride deposition rate (mg/m^2.d)	Chloride class according to ISO 9223	Sulphur dioxide concentration level (μg/cm^3)	SO_2 class according to ISO 9223	Time of wetness per year (hrs)	Time of wetness class according to ISO 9223
Port Louis	6.9	S_1	13.1	P_1	1897	T_3
St Julien d'Hotman	2.3	S_0	> 2.6	P_0	6651	T_5
Belle Mare	5.3	S_1	> 2.6	P_0	3621	T_4
Reduit	5.8	S_1	14.7	P_1	5863	T_5

From the corrosion loss trend obtained over the period of exposure, the corrosivity of the atmosphere at each location was determined according to ISO 9223 (2012), and the corrosion behaviour of the steel was monitored and explained.

During the period that the metal samples were exposed, the average level of airborne salinity, sulphur dioxide level and TOW also were determined. Airborne salinity was monitored twice yearly, during the summer and winter months, using the wet candle method. Sulphur dioxide levels were measured from air quality monitoring stations from the Ministry of Environment (Mauritius), and the TOW was determined from the relative humidity and temperature values obtained from meteorological stations at the sites.

Results and analysis

Atmospheric parameters

The average values of the atmospheric parameters monitored are shown in Table II. They have been classified according to ISO 9223. It can be observed that although Mauritius is a small island, the level of airborne salinity is quite low. This is most probably due to the fact that the coral reefs are very far from the coast and, therefore, the sea in Mauritius is normally very calm. Pollution levels are also low in Mauritius. The TOW, on the other hand, is very high, especially in regions of St Julien d'Hotman and Reduit.

Mass loss analysis

The results of the mass loss analysis are shown in Table III.

It is widely accepted that long-term atmospheric corrosion of steel conforms to an equation of the form (Wei, 1991; Morcillo *et al.*, 1993):

$$C = At^B \tag{1}$$

where C is the corrosion loss in (μm) and t is the time (months). A is generally considered as a measure of the initial reactivity of the material with the environment and it is numerically equivalent to the corrosion loss when time is unity. The exponent B is in turn a function of the type of atmosphere where the metal is exposed. In other studies, t has been taken as the TOW in the period under consideration (Veleva and Maldonado, 1998).

It is well-known that the corrosion of steel in the atmosphere follows a mechanism controlled by the diffusion of reactants across the rust layer (Benarie and Lipfert, 1986). The value of B, in equation (1), close to 0.5, results from an

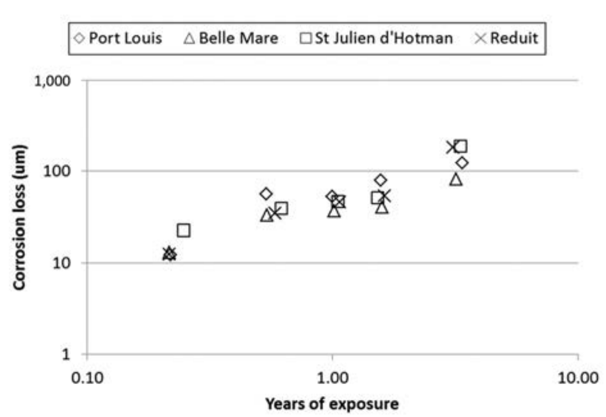
Table III Results of mass loss analysis

Site	Removal	Years of exposure	Average corrosion loss (μm/year)
Port Louis	1	0.22	12.30
	2	0.54	56.58
	3	1.00	52.55
	4	1.57	79.12
	5	3.37	123.80
St Julien d'Hotman	1	0.25	22.84
	2	0.62	39.35
	3	1.05	46.34
	4	1.52	51.66
	5	3.33	187.57
Belle Mare	1	0.22	13.16
	2	0.54	32.90
	3	1.01	36.82
	4	1.59	40.66
	5	3.19	81.55
Reduit	1	0.22	12.46
	2	0.58	34.47
	3	1.07	46.19
	4	1.63	53.79
	5	3.10	181.54

Table IV Equation for corrosion loss

Site	Equation	R^2
Belle Mare	$C = 37.3t^{0.61}$	0.93
Port Louis	$C = 55.3t^{0.78}$	0.87
Reduit	$C = 48.7t^{0.91}$	0.94
St Julien d'Hotman	$C = 54.3t^{0.74}$	0.87

Figure 4 Log-log plot of corrosion loss against years of exposure



ideal diffusion-controlled mechanism. This situation seems to occur in slightly polluted inland atmospheres (Morcillo *et al.*, 1993). The value of B is greater than 0.5 when there is an acceleration in the diffusion process, for example due to cracking or flaking of the rust layer. The value of B smaller than 0.5 would result from a decrease in the diffusion coefficient with time through compaction of the rust layer, recrystallization, etc.

For the corrosion loss results, the bilogarithmic models of the corrosion losses for each of the sites considered are shown in Table IV. The coefficient of determination (R^2) varies between 0.87 and 0.94. The bilogarithmic law explains well the corrosion trend for Belle Mare and Port Louis, as shown in Figure 4.

The situation is different for St Julien d'Hotman and Reduit. The points, being proportionally scattered around the trend-curve, result in a high coefficient of determination. However, the trend is such that the corrosion rate is high initially, for a period of six months and then decreases significantly for a period of one year and eventually increases again, as shown in Figure 5. This trend has been observed in tropical marine environments (Ma *et al.*, 2010) and is known

Figure 5 Variation of corrosion loss with time at Reduit and St Julien d'Hotman

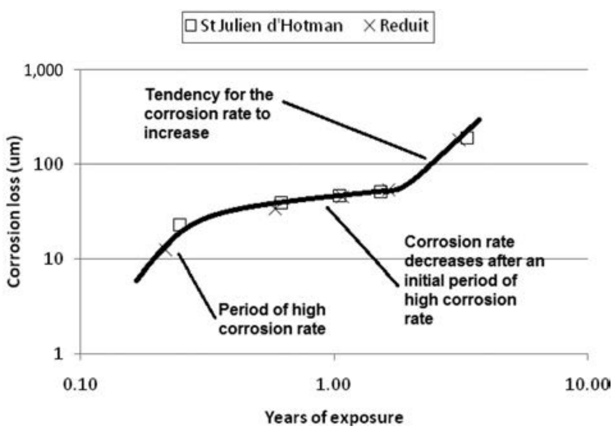
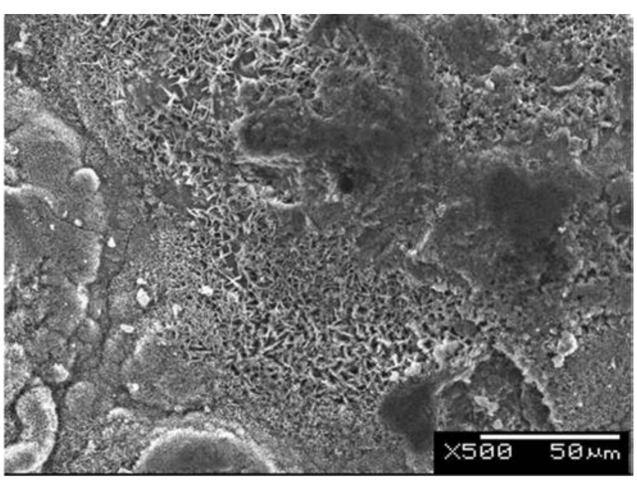


Figure 6 SEM micrograph for the first removal for the site at Port Louis



as the "Reverse Phenomenon". This type of behaviour has been observed in some sites having high airborne salinity level. In this study, this type of behaviour was observed at the sites having the highest TOW. In fact, TOW was found to be a main parameter affecting atmospheric corrosion and it has even been used in atmospheric corrosion modelling (Guttman and Sereda, 1968).

SEM and Raman tests

Considering the sites of Port Louis and Belle Mare

A high initial corrosion rate was observed through the mass loss analysis, which can be explained by the presence of a porous rust layer on the base metal, as shown in Figure 6. Sandy crystals (Antunes *et al.*, 2003) and flowery structures of fine plates (de la Fuente *et al.*, 2011) of lepidocrocite could be identified on the rust surfaces, leading to an irregular and open

Figure 7 SEM micrograph for the third removal for Port Louis

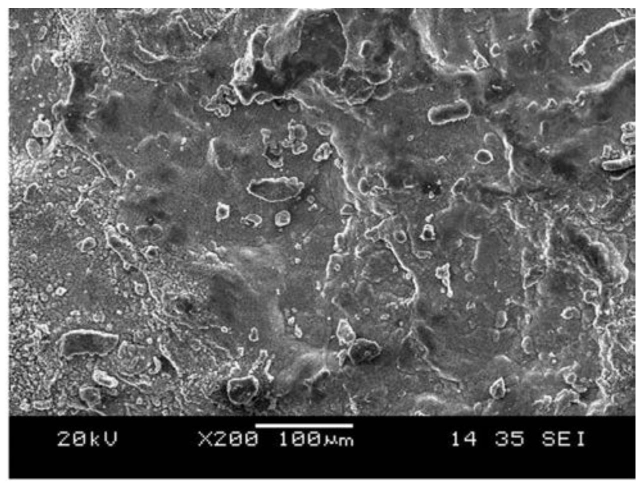
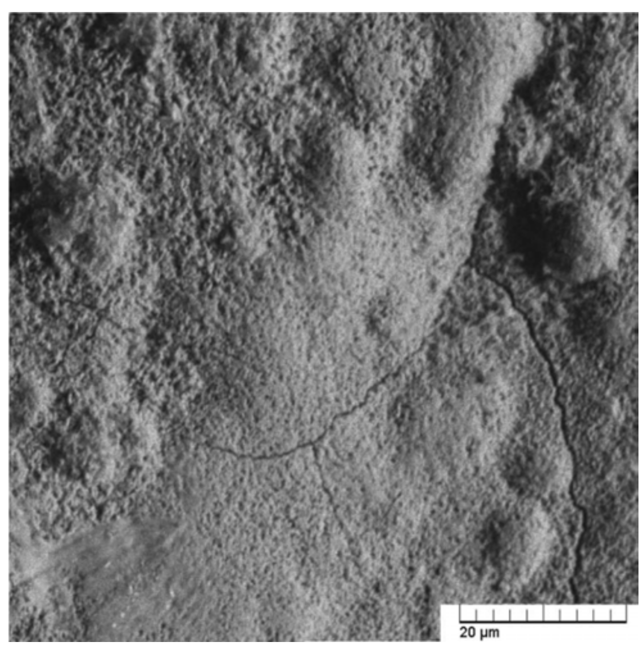


Figure 8 SEM micrograph for the fifth removal for Port Louis



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structure that permitted easy access of corrosive species to the underlying metal. This morphology also was observed for the third and fifth removals, as shown in Figures 7 and 8, respectively. In fact, flaking and cracking was observed on the surface, which could have counteracted the effect of the increase in the thickness of the rust layer. For the fifth removal, apart from the cracks, corrosion nests were observed on the rust surface. This ensured that the corrosion proceeded at a significant rate despite the increase in thickness of the rust layer.

Considering Reduit and St Julien d'Hotman

For these sites, the initial rust layer was similar to that obtained at Port Louis and Belle Mare. However, from 6 months up to 18 months of exposure, there was a significant decrease in the corrosion rate. From the SEM micrographs, the formation of magnetite as a dark and compact layer (Xiao *et al.*, 2008) could be observed, as shown in Figure 9.

Figure 9 SEM micrograph for the third removal for Reduit

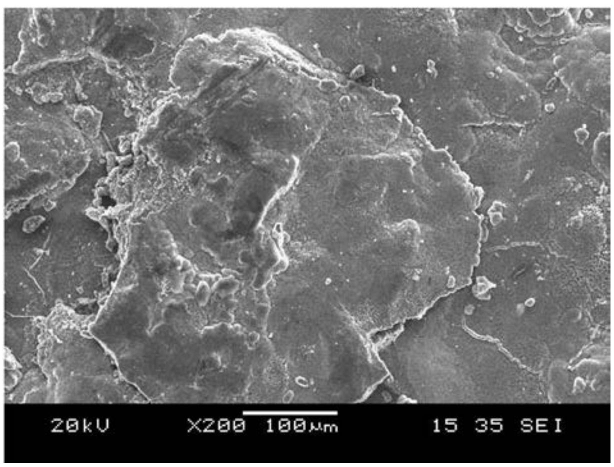
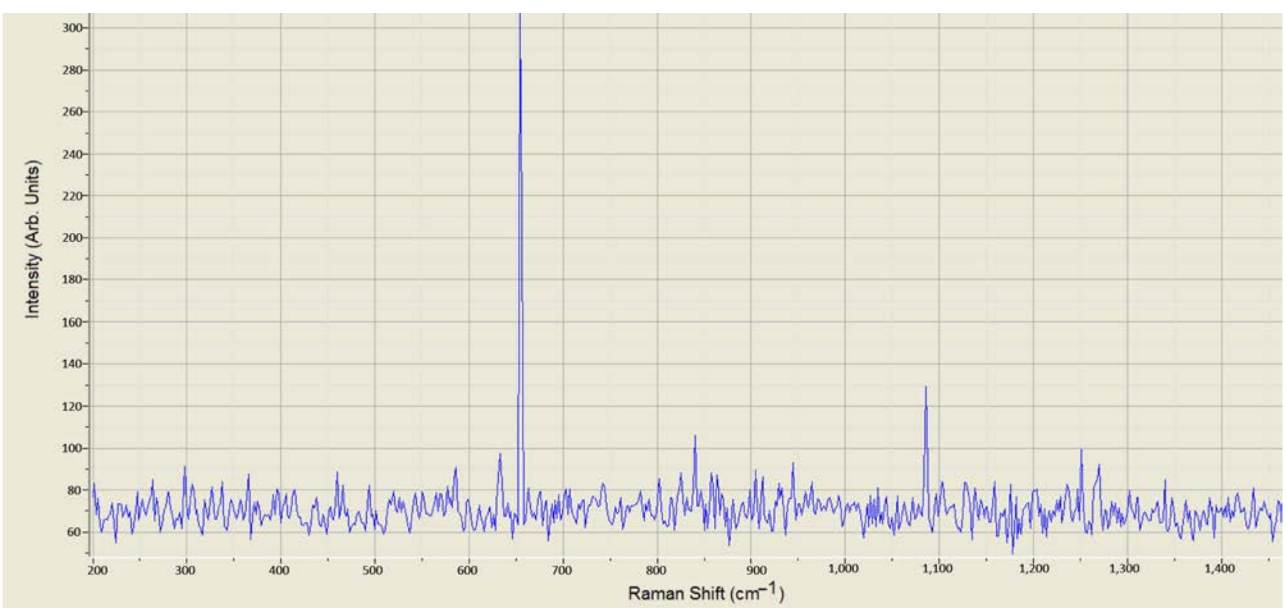


Figure 10 Raman tests for the third removal for Reduit



Although magnetite is a good conductor, it is considered protective because of its compactness and thermodynamic stability (Hoerle *et al.*, 2004). This would clearly lead to a substantial decrease in the corrosion rate. Its presence suggests that a change in the atmospheric corrosion mechanism took place, as compared to the first six months of exposure. Raman spectroscopy results also confirm the presence of magnetite in the rust layer, with a peak at 660 cm^{-1}, as shown in Figure 10. After 18 months of exposure, cracking and flaking of the rust layer was observed, as shown in Figure 11, leading to a significant increase in corrosion rate. A mixture of phases was observed on the rust surface. This was confirmed from the Raman test results, as shown in Figure 12.

Corrosivity

The corrosion loss over the first year is given in Table V. According to ISO 9223, the corrosivity of the Mauritian atmosphere lays in category C3 and C4. Belle Mare is a marine site but had the lowest corrosion loss. Sites which are inland, such as Reduit and St Julien d'Hotman, had much higher corrosion losses. Hence, it could be concluded that nearness to the sea was not a major factor affecting atmospheric corrosion in Mauritius.

As expected, Port Louis, which is a marine and industrial site, had the highest corrosivity. The corrosivity of Reduit was very near to the C3-C4 borderline. Whereas, St Julien d'Hotman, which is a rural site, had the second highest corrosion loss. St Julien d'Hotman has a humid atmosphere with high TOW. Considering the cases of Reduit and St Julien d'Hotman, it can concluded that TOW can be considered as a major factor affecting atmospheric corrosion in the inland region.

The corrosivity of the Mauritian atmosphere, therefore, can be regarded as being moderate to severe. Under these circumstances, it is essential to use proper corrosion

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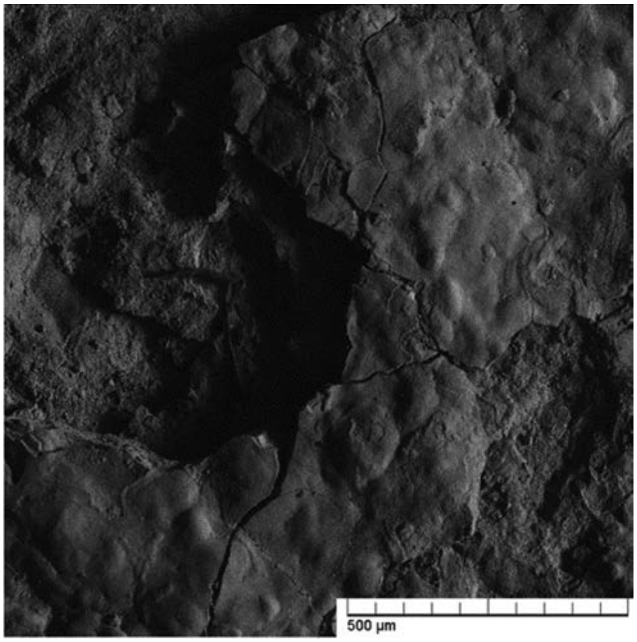
Figure 11 Fifth removal for Reduit



prevention methods when the steel structures are erected outdoors. The harbour area, which is situated in Port Louis, is the most affected region in the island. In this area, certain companies make use of chlorinated rubber and epoxy paints for atmospheric corrosion prevention. Many companies, however, use alkyd-based paints, mainly due to economic factors (Surnam, 2013). This may, however, prove to be costly because a failure in the paint system can lead to rapid corrosion loss. For large structures, such as telecommunication towers, epoxy coatings are used (Surnam, 2013). They are frequently inspected and maintained to ensure that there is no sign of corrosion on them. Galvanising is another corrosion prevention method that is commonly used in Mauritius. Hotels located in the coastal areas use alternative materials, such as aluminium, wood and PVC, instead of carbon steel. This is, however, used mainly because of the better aesthetic value of these materials (Surnam, 2013). There are also some companies which do not use any corrosion prevention methods. However, taking into consideration the corrosivity of the Mauritian atmosphere, this type of strategy may prove to be very costly in the long term.

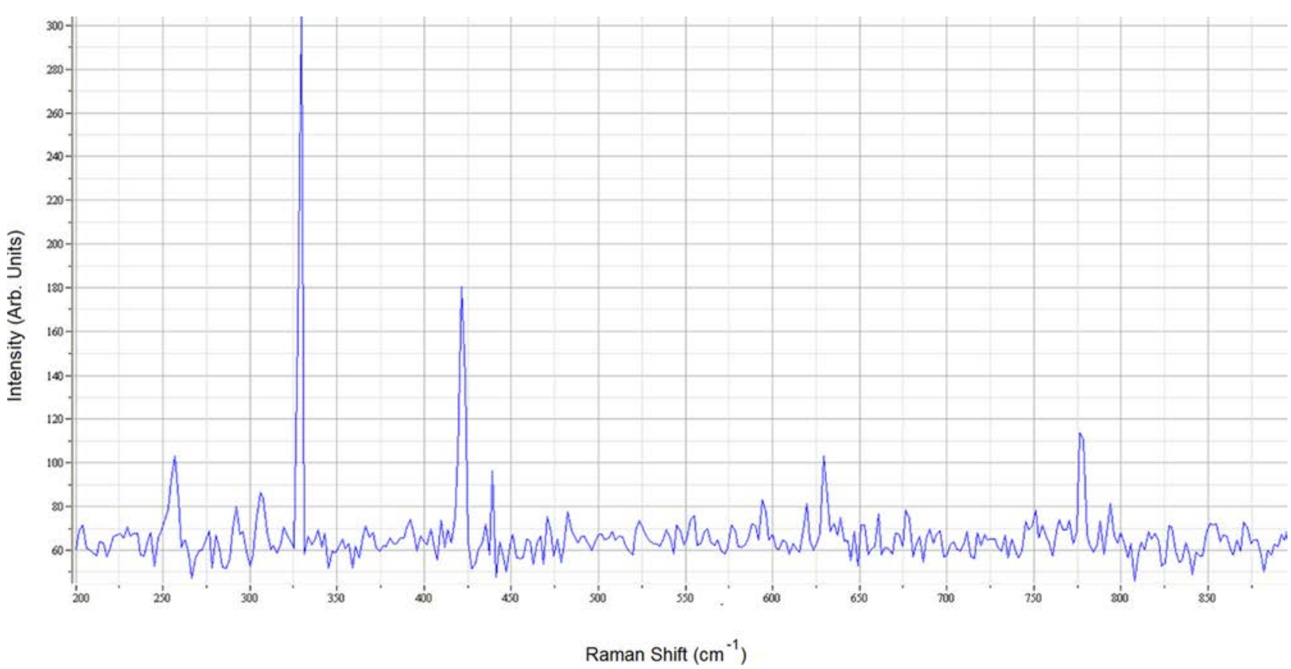
Table V Corrosivity of the atmospheres in Mauritius

Site	Corrosion loss in first year (μm)	Corrosivity category
Belle Mare	37.3	C3
Port Louis	55.3	C4
Reduit	48.7	C3
St Julien d'Hotman	54.3	C4

Conclusion

Out of the four sites considered in Mauritius, it was observed that two of them did not clearly corrode according to the bilogarithmic law. The two sites, namely, St Julien d'Hotman and Reduit, showed the "reverse phenomenon" behaviour. The corrosion behaviour of carbon steel at all the sites was successfully explained through SEM and Raman tests. Moreover, it was observed that TOW was an important parameter affecting atmospheric corrosion of the steel in the inland region. The corrosivity of the atmosphere in Mauritius was found to vary between corrosivity categories C3 and C4, based on ISO 9223, which refer to moderate to severe atmospheres.

Figure 12 Raman test results for the fifth removal at Reduit



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Corresponding author

Baboo Y.R. Surnam can be contacted at: y.surnam@uom.ac.mu